



Examiners' Report Lead Examiner Feedback

June 2022

Pearson BTEC Nationals
In Applied Science (31617H)
Unit 1: Principles and Applications of Science I

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this [link](#)

Awarding BTEC qualifications in 2022

Ofqual has [set out their plans](#) for awarding qualifications in 2022 and intend to return to a normal, pre-pandemic, approach to grading standards over by 2023. They have confirmed that 2022 will be a transition year, to reflect that we are in a pandemic recovery period and students' education has been disrupted.

Our guiding principle and approach to awarding BTEC qualification results in 2022 will be to ensure parity in relation to the approach being taken for GCSE and A level learners. BTEC courses have a different structure and design to academic qualifications - BTECs are modular qualifications (with assessments taking place throughout the course) compared to GCSEs and A levels which

are linear (assessed and awarded at the same time at the end of the year), and therefore our approach needs to be different.

In 2022 we will return to the usual method of calculating BTEC qualification results, however adaptations including, U-TAGs and reduced internal assessment, are in place to provide a comprehensive package of support for students.

The basis of our awarding approach to BTECs this year is to ensure it is as fair as possible for all learners. We will use a range of evidence to set grade boundaries for the external units. Part of this evidence will be to closely monitor learner performance in all assessments that contribute to learners' final qualification grade, to ensure parity with A level and GCSEs.

Further information can be found on our website and via our Social Media channels.

Introduction

Biology

Structure and functions of cells and tissues.

Medical professionals need to understand the structure and workings of cells. They build on this knowledge to understand how the body stays healthy as well as the symptoms and causes of some diseases. This allows them to diagnose and treat illnesses. The study of bacterial prokaryotic cells gives an understanding of how some other diseases are caused and can be treated.

Introduction to the Overall Performance of the Unit

Structure and functions of cells and tissues.

This examination of the Level 3 Applied Science Unit 1 Biology section was well received by learners, with many making good attempts at all questions and fewer leaving blank spaces than in previous series. Exam technique and the understanding of the command verbs is continuing to improve. It is beneficial for learners to appreciate the requirement of each of the command verbs so that they can target their response and provide appropriate credit worthy answers. Explain is the command verb which seems least understood, especially for a 2 mark question. Learners tend to make an identification point without a linked expansion for full marks, or learners give a number of points, suggesting an imbalance of time spent on short questions.

Learners should show that they understand the relationship between the structure and function of cells and tissues. Learners still struggle to recall definitions and find it difficult to provide links between different related aspects of the specification.

Introduction

Chemistry

This was the second sitting of the Level 3 Applied Science Unit 1 Chemistry section since the pandemic. It is clear that centres are ensuring that their learners are aware of the content being tested using the additional guidance document and practicing questions using the past papers available. This approach with learners has meant that fewer blank spaces were seen and many more learners were achieving marks across the whole of the paper.

Introduction to the Overall Performance of the Unit

Learners that were successful this year were able to perform the given calculations using clear and logical working. They were able to understand and use scientific vocabulary in the correct context and apply their knowledge of chemistry explain trends in data and ideas. Their knowledge of basic scientific terminology allowed them to access questions and apply ideas.

Introduction

Physics

This examination allows learners to show their knowledge and understanding of the features of all waves with particular emphasis on those waves that form part of the electromagnetic spectrum and the uses and application of these waves to communication.

Introduction to the Overall Performance of the Unit

Learners were able to demonstrate a much improved understanding of the use of line spectra to identify elements in stars although knowing how water particles move in a water wave was found to be more challenging.

Learners also found it more difficult to identify the period of a wave than to select the number of oscillations shown on a graph. Given a simple equation the majority of learners were able to rearrange the equation to find the subject and calculate the correct answer. However the use of $\sin$ and $\sin^{-1}$ still presents a challenge to many learners and needs more practice.

Describing an analogue signal still presented difficulties for some learners and the relationship between the graphical representations of the analogue signal converted to digital signals with different sampling rates was not appreciated by many learners.

The majority of learners were able to access the extended open response question and provide at least one comparison between the waves used to transmit signals through optical fibres and the waves used to transmit signals to satellites. Some learners had sufficient relevant knowledge and understanding to support the comparison with a clear, coherent and logical chain of reasoning.

Individual Questions

Biology

Q1(a)(i)

This question was answered well by the majority of learners. Learners were able to identify structure X as the capsule. The most common incorrect response was B cell wall.

(a) (i) Identify structure X in Figure 1.

- ☒ A capsule
- ☒ B cell wall
- ☒ C plasmid
- ☒ D ribosome

Q1(a)(ii)

This question was answered well by the majority of learners. Most learners gave the idea of DNA or genetic material, but often tried to describe the DNA as a stand, free flowing, etc. A common incorrect response was plasmid.

Lead Examiner comment: *This response was awarded 1 mark. The learner has given DNA, which is a correct response. We can accept DNA strand, DNA string, genetic material, chromosome, nucleic acid.*

(ii) The nucleoid is made of structure Y.

Identify structure Y in Figure 1.

(1)

DNA Strand

Lead Examiner comment: *This response was awarded 0 marks.*

'Plasmid(s)' was a common incorrect response. Unfortunately, in this example the correct response (DNA) has been given but crossed out and replaced, and so the replacement has to be accepted as the learner's response.

(ii) The nucleoid is made of structure Y.

Identify structure Y in Figure 1.

(1)

~~DNA~~ plasmids

Q1(a)(iii)

This question was answered well by the majority of learners.

Most learners were able to mention the cell membrane controlling the movement of substances into and out of the cell. Some learners structured their response using a named substance.

Although it is an acceptable response, there were very few references to cell signalling. MP2 was awarded most often. Learners would often make statements like 'keep things in' or 'allowing things in and out'.

Some learners gave incorrect responses confusing the plasma membrane with a cell wall or nuclear membrane, referring to protection, structure or shape, or fighting off pathogens.

Lead Examiner comment: *This response was awarded 1 mark. This is correct phrasing for MP2.*

(iii) State **one** function of the plasma membrane.

(1)

controls permeability, what can enter or leave the cell.

Lead Examiner comment: *This response was awarded 0 marks.*

'Protects' is a common incorrect response and suggests a confusion with the cell wall, rather than the plasma membrane. It is too vague to be creditworthy.

(iii) State **one** function of the plasma membrane.

(1)

Protects everything inside it

Q1(b)

This question was not well answered by the majority of learners.

The most frequently occurring correct response was regarding the presence of a membrane/membrane bound nucleus or envelope. Some learners were also able to identify the presence of the nucleoplasm/nucleolus and chromosomes.

The most common misconception was that only the nucleus contains genetic material. A large number of learners obviously did not know what the nucleoid was and gave DNA or genetic information.

Some learners may have been confused over 'feature' and 'function'. Several learners misinterpreted the question and provided responses regarding the nucleus controlling the cell/controlling the cells activities. There were also quite a few references to the shape of the nucleus and nucleoid.

Lead Examiner comment: *This response was awarded 1 mark. There is a long list of possible features- the learner only has to give 1.*

(b) Eukaryotic cells contain a nucleus instead of a nucleoid.

Give **one** feature of the nucleus that is **NOT** found in the nucleoid.

(1)

in eukaryotic cells it contains a nucleolus

Lead Examiner comment: *This response was awarded 0 marks. The response is incorrect; both contain DNA.*

(b) Eukaryotic cells contain a nucleus instead of a nucleoid.

Give **one** feature of the nucleus that is **NOT** found in the nucleoid.

(1)

a nucleus holds dna whereas nucleoids don't.

Q2(a)

This question was answered well by the majority of learners.

Most learners were able to correctly select aerobic respiration.

(a) Identify which process happens inside mitochondria.

- ☒ A aerobic respiration
- ☒ B crossbridge formation <
- ☒ C fertilisation >
- ☒ D phagocytosis x

Q2(b)

This question was answered well by the majority of learners. Most learners were able to correctly complete the table with yes or no.

Some learners gave incorrect responses for amyloplasts, indicating that they would be present in animal and not plant cells. Some learners also indicated that ribosomes are not found in plant cells. Knowledge of cytoplasm in both plant and animal cells was consistent for most learners.

Lead Examiner comment: *This response was awarded 3 marks.*

(b) Complete Table 1 to show which organelles are found in animal cells and which organelles are found in plant cells.

The first row has been completed for you.

(3)

organelle	found in animal cells	found in plant cells
cell wall	No	Yes
ribosomes	Yes	Yes
cytoplasm	Yes	Yes
amyloplasts	No	Yes

Table 1

Q2(c)

This question was answered well, with the majority of learners gaining at least 1 mark. Many answers were awarded a mark for an identification point but learners failed to go on to gain the expansion mark, suggesting a lack of understanding in the command verb.

Most learners referred to the presence of chloroplasts/chlorophyll and their role in trapping light energy from the sun. There were also several responses regarding the large vacuole and its role in pushing the chloroplasts to the perimeter of the palisade cells. There were also several comments regarding the large surface area and how this aids the absorption of light energy.

There were very few responses that included information regarding the cylindrical shape of the cells, or the role of the cytoskeleton.

Lead Examiner comment: *This response was awarded 2 marks.*

The learner has given a line of explanation which is creditworthy for 2 marks. The learner has given the identification point of 'contains lots of chloroplasts'. Their comment about chlorophyll is also creditworthy for the same marking point. The link to 'attracts the sunlight' is creditworthy for MP2 and so 2 marks can be given.

(c) A palisade mesophyll cell is a specialised plant cell.

Explain **one** feature of a palisade mesophyll cell that helps it to photosynthesise.

(2)

*It has a palisade layer of cells near to
it contains lots of chloroplasts which
contain lots of chlorophyll which is the
green pigment within a plant cell which
also attracts the sunlight during photosynthesis.*

(Total for Question 2 = 6 marks)

Lead Examiner comment: *This response was awarded 0 marks.*

This answer is too vague as it could be a confusion with respiration.

(c) A palisade mesophyll cell is a specialised plant cell.

Explain **one** feature of a palisade mesophyll cell that helps it to photosynthesise.

(2)

*it has a lot of ways oxygen can get
through.*

Q3(a)

This question was answered well by the majority of learners.

The vast majority of learners were able to give a function that indicated movement. Swim was an accepted phrasing.

The most common incorrect response stemmed from a misreading of the question and consequently naming structure Z instead of giving its function.

Lead Examiner comment: *This response was awarded 1 mark. We can accept the idea of move or swim. The learner does not need to name structure Z.*

3 Egg cells and sperm cells are two specialised cells found in humans.

Figure 2 shows a diagram of an egg cell and a sperm cell. They are not drawn to scale.

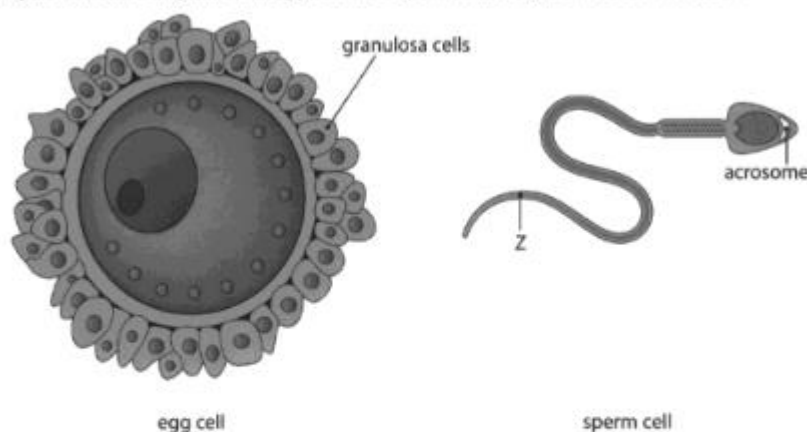


Figure 2

(a) State the function of the structure labelled Z in Figure 2.

(1)

The Flagellum which is what makes it move / swim. ~~move~~

Q3(b)

This question was not well answered.

The most common mark was for giving 'enzymes' for T. Some learners gave the correct responses but in the wrong order, but the correct words for R and S were rarely seen.

Lead Examiner comment: *This response was awarded 3 marks. Although the spelling for R and S are not correct, we can give benefit of doubt for phonetic spellings and so the marks have been awarded.*

(b) Paragraph 1 describes the role of the sperm cell head in penetrating the egg cell during fertilisation.

The sperm cell penetrates the granulosa cells of the R
The head of the sperm cell binds with the S of the egg cell.
The sperm head releases T to digest the glycoprotein protective coat of the egg cell.

Paragraph 1

Identify the missing words R, S and T in Paragraph 1.

(3)

R coronach radiata
S ^{membrane} zona pecuda
T ~~acrosome~~ ~~enzyme~~ enzymes

Lead Examiner comment: *This response was awarded 1 mark.*

We cannot credit R as egg as this would be a repeat of the stem above Paragraph 1. It was a common insufficient response and does not gain a mark. The spelling for S is too far from the correct response and so cannot be given credit. A mark can be awarded for "enzymes" for T.

(b) Paragraph 1 describes the role of the sperm cell head in penetrating the egg cell during fertilisation.

The sperm cell penetrates the granulosa cells of the R
The head of the sperm cell binds with the S of the egg cell.
The sperm head releases T to digest the glycoprotein protective coat of the egg cell.

Paragraph 1

Identify the missing words R, S and T in Paragraph 1.

(3)

R egg cell.
S Zona
T enzymes from the acrosome

Q3(c)

This question was answered well by some learners.

Learners could give one of several possible responses, such as haploid, half the number of chromosomes, or 23 chromosomes. A comparison was not required as this was a 1 mark question.

A common incorrect response was the vague idea about the size of the nucleus, with learners saying that the epithelial cell nucleus was bigger. Another common misconception was that the epithelial cell does not contain genetic material.

Lead Examiner comment: *This response was awarded 1 mark.*

Just stating that the sperm cell is haploid is sufficient alone; the learner does not need to also state that the epithelial cell is diploid for the mark and we can give BOD that being haploid is a difference. The learner has also stated that the sperm cell "only contains half of the genetic material" which would have been creditworthy on its own.

(c) A sperm cell and an epithelial cell each contain a nucleus.

State **one** difference between the nucleus of a sperm cell and the nucleus of an epithelial cell.

(1)

a sperm cell is a haploid cell so only contains half of the genetic material on epithelial cell is a ~~diploid~~ ^{diploid}.

Lead Examiner comment: *This response was awarded 0 marks.*

The learner is incorrect about the epithelial cell. Just containing genetic information is not a difference without details and so this is not creditworthy.

(c) A sperm cell and an epithelial cell each contain a nucleus.

State **one** difference between the nucleus of a sperm cell and the nucleus of an epithelial cell. (2 marks)

(1)

the nucleus of the sperm cell contains genetic information, epithelial cell's nucleus does not.

(Total for Question 3 = 5 marks)

Q4(a)

This question was answered well by the majority of learners. Smoking was a very common correct response but of the list of other possible responses, most were rarely seen. The cause of COPD was sometimes confused with CVD and atherosclerosis and so some learners incorrectly suggested that an unhealthy diet would be a cause.

Lead Examiner comment: *This response was awarded 1 mark. Smoking was a very common, correct response.*

4 Chronic obstructive pulmonary disease (COPD) is a common disease in the UK.

(a) Give **one** cause of COPD.

(1)

Smoking

Lead Examiner comment: *This response was awarded 0 marks. Reference to diet is incorrect as a cause of COPD.*

4 Chronic obstructive pulmonary disease (COPD) is a common disease in the UK.

(a) Give **one** cause of COPD.

(1)

unhealthy diet

Q4(b)

This question was answered well by learners, with most learners gaining at least 2 out of the 4 marks. Learners were able to achieve MP1 and MP2 by taking the correct reading and calculating the difference.

Often learners would start with one method and then go over to the alternative method. Some learners who chose to use the alternative method forgot to subtract 100 from their answer and this limited their overall marks to 3.

Common errors were incorrect reading from graph (reading 3500 instead of 3750), dividing by wrong total (method 1), using an upside-down division (method 2), or forgetting to multiply by 100.

Particularly common was confusing 'per 100,000' as meaning that 100,000 needed to be the divisor. Also common was calculating the % of the total population with COPD at the start (3%) and the % of total population with COPD at the end (3.75%).

Lead Examiner comment: *This response was awarded 4 marks. 25% is the correct response and so 4 marks are awarded.*

(b) Figure 3 shows the number of people, per 100,000 population, diagnosed with COPD from 2004 to 2012 in the UK.

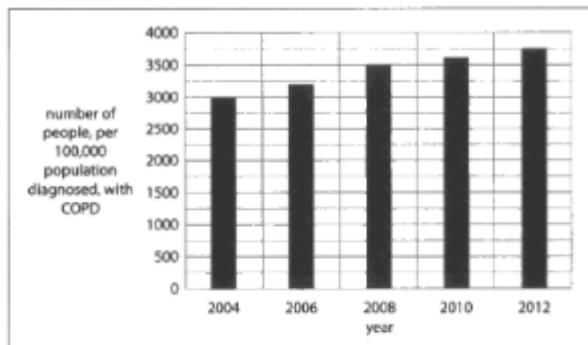


Figure 3

Calculate the percentage increase in the number of people diagnosed with COPD, per 100,000 population, from 2004 to 2012.

Use the information in Figure 3.

(4)

Show your working.

$$25\% \text{ OF } 3000 = 3000 \div 4 = 750$$

$$3000 + 750 = 3750 = 25\% \text{ increase}$$

$$\text{percentage increase} = 25\% \%$$

Lead Examiner comment: *This response was awarded 2 marks.*

The response is incorrect and so the working should be investigated for marks and ECF. The learner has given the values needed for the calculation from the graph (3750 and 3000) and so MP1 can be given.

The difference between the values the learner has taken from the graph has been given for MP2. The learner has then divided by 100 which is not a relevant step for either method. Overall, 2 marks can be awarded.

(b) Figure 3 shows the number of people, per 100,000 population, diagnosed with COPD from 2004 to 2012 in the UK.

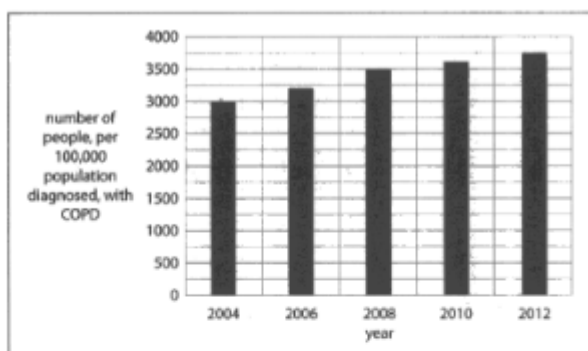


Figure 3

Calculate the percentage increase in the number of people diagnosed with COPD, per 100,000 population, from 2004 to 2012.

Use the information in Figure 3.

(4)

Show your working.

$$\begin{aligned} 3750 - 3000 \\ = 750 \div 100 \\ = 7.5\% \end{aligned}$$

$$\begin{aligned} 3000 \text{ in } 2004 \\ 3750 \text{ in } 2012 \end{aligned}$$

percentage increase = 7.5 %

Q4(c)

This question was answered well by the majority of learners with most achieving no more than 2 marks.

Most learners were able to connect less/damaged alveoli to a lower surface area, with 'build-up of mucus' being the second most common response. Learners often described identification points but lacked an appropriate expansion so failed to get full marks while others appear to have become confused and talked about blood pressure, blocked arteries and blood vessels being damaged.

There were often references to the causes of COPD, eg, tar build-up, dust, particles, smoking, coughing, damaged cilia and airways being blocked. Generic points such as 'lung damage' were often seen, as were generic descriptions of gas exchange.

References to macrophages or enzyme damage were rarely seen and many learners repeated the stem of the question in their answers. A lot of answers were an attempt to interpret the photographs without any understanding of the disease process.

Lead Examiner comment: *This response was awarded 3 marks.*

The learner has given some details we can ignore at the start of the response. The comment about scarring and thickness of the walls is the same marking point in this context (MP10). Careful not to double credit the same marking point with different phrasing. The comment about reduces elasticity (MP5) and "inflate and deflate" is sufficient for alveoli can't recoil (MP6).

As any marking point could be an identification point or an expansion point depending on how the learner shapes their response it is acceptable to link MP5 and MP6. Overall, 3 marks can be awarded.

(c) Figure 4a shows a photomicrograph of alveoli from a healthy person.

Figure 4b shows a photomicrograph of alveoli from a person with COPD.

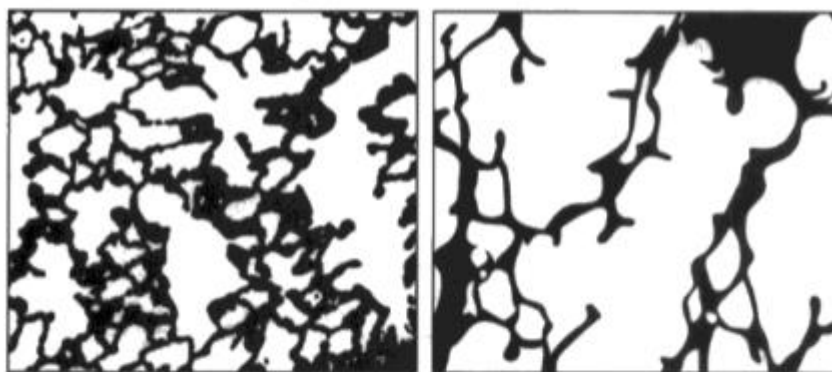


Figure 4a

Figure 4b

COPD reduces the efficiency of gas exchange in the lungs.

Explain why.

(4)

- This is because one of the causes of COPD is smoking. This ~~then causes inflammation~~ ^{irritates, inflames} and causes scarring of the alveoli walls.
- Due to the excessive scarring on the walls of the alveoli it increases the thickness of the walls which affects the efficiency of gaseous exchange.
- The scarring also reduces the alveoli's elasticity so it can't inflate and deflate.

Lead Examiner comment: *This response was awarded 0 marks.*

"Not much space" is not sufficient for reduced surface area.

The details are not sufficient for the marking points about the concentration gradients, they have just repeated the stem and so no marks can be awarded.

(c) Figure 4a shows a photomicrograph of alveoli from a healthy person.

Figure 4b shows a photomicrograph of alveoli from a person with COPD.

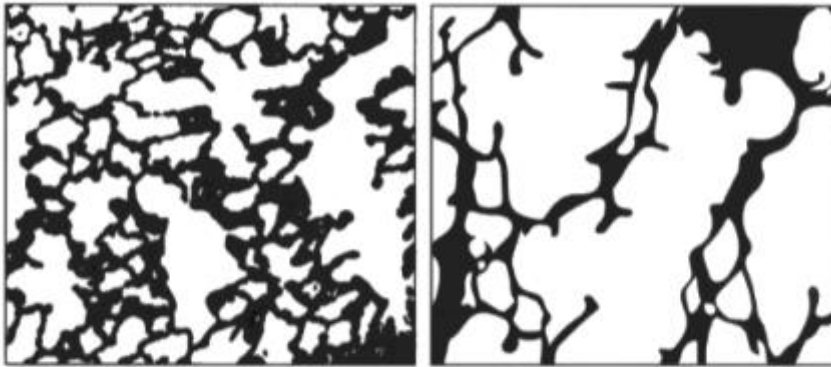


Figure 4a

Figure 4b

COPD reduces the efficiency of gas exchange in the lungs.

Explain why.

(4)

there is not much space so the
gas to get trapped in figure 4b
therefore the exchange of gas would
be at a low.

Q5

This question was answered well by the majority of learners, however, many learners failed to enter Level 3 as they did not explain their answers and only provided a basic understanding. Many learners thought non-myelinated were faster.

References to speed and insulation most commonly seen, as was the idea of jumping across nodes of Ranvier. Disappointingly, a relatively large number appear to have looked at the diagrams and concluded that the myelin sheath acted as a barrier to conduction and slowed it down. Good answers discussed insulation/reducing loss of ions, saltatory conduction (or at least 'jumping' in some form) and used correct terminology.

There was some confusing of synapses and Nodes of Ranvier, and of Schwann cells with axons. Confusion between electrical and thermal insulation was also common, as was a general misunderstanding regarding muscle cells (slow/fast twitch etc.).

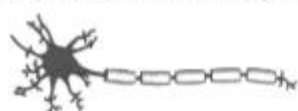
Lead Examiner comment: This response was awarded 6 marks.

The learner has given a plan at the start of their response which we can consider. They have then gone on to provide a discussion of both myelinated and non-myelinated neurones.

They have been concise, without any irrelevance, and so comprehensive knowledge and understanding is shown. The response is well structured. The learner does not have to cover every point in the indicative content to be awarded maximum marks- this meets the criteria for the Band 3 level descriptor.

5 Figure 5a shows the structure of a myelinated neurone.

Figure 5b shows the structure of a non-myelinated neurone.



myelinated neurone

Figure 5a



non-myelinated neurone

Figure 5b

Discuss the reasons for the differences in the speed of impulse conduction along myelinated and non-myelinated axons.

(6)

Non-myelinated neurone	Myelinated neurone
- Gray - Slow impulses - doesn't have myelin sheath - No nodes of Ranvier	- white - fast impulses - Does have myelin sheath - Does have nodes of Ranvier

There's a difference in the speed of impulse conduction between myelinated & non-myelinated neurones. The speed of impulse conduction in myelinated neurones is faster than non-myelinated neurones. The myelin sheath layer allows the impulse to jump from one node of Ranvier to another allowing the impulse to travel quicker, fast.

~~Whereas non-myelinated~~
However the lack of myelin sheath prevents the impulse to travel fast, this is because as the impulse travels along the neurone it's consistently going through action potential and resting potential therefore it will take time for ^{the} impulse to travel to the end of the neurone. (Total for Question 5 = 6 marks)

However ⁱⁿ myelinated neurones the impulse jumps from one node to another ~~per~~ due to the myelin sheath covering large portion of the surface of the neurone, therefore the impulse goes through action potential ~~resting potential through the action~~ of jumping at small gaps ~~called~~ ^{which is known} as nodes of Ranvier by jumping, quickly reaching to the end of the neurone.

Lead Examiner comment: *This response was awarded 3 marks.*

The learner has shown good, but not comprehensive knowledge. The learner has made the same point over again in a number of ways.

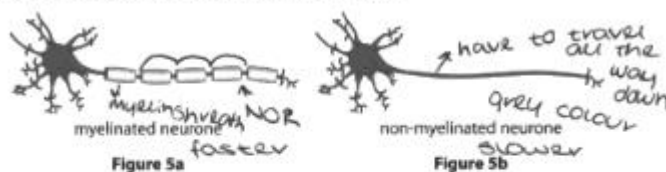
The learner has given an idea of protection, rather than insulation. There is no reference to ions or lipids.

This is a Band 2 response- this is borderline 3 to 4 marks.

There are some key details missing, as the steps in the process are not complete, and so 3 marks have been given. The lines of argument are not supported sufficiently.

5 Figure 5a shows the structure of a myelinated neurone.

Figure 5b shows the structure of a non-myelinated neurone.



Discuss the reasons for the differences in the speed of impulse conduction along myelinated and non-myelinated axons.

(6)

- in a myelinated neurone the nerve impulses will travel faster
- in a non myelinated neurone impulses will travel slower
- The myelinated neurone contain Schwann cells protected by myelin sheath on the axon
- inbetween the cells there are little gaps called nodes of ranvier which allow nerve impulses to jump between ^{the cells} allowing the nerve impulse to travel faster.
- on a non myelinated neurone they don't have the nodes of ranvier meaning the impulse will have to travel along the whole axon causing the impulse to take longer
- The myelin sheath on the myelin-

ated neurone also is able to protect the neurone from damage allowing impulses to travel smoothly.
-non myelinated neurones don't have myelin sheath to protect the axon when impulses travel down.
-Myelinated neurone overall have faster nerve impulses (Total for Question 5 = 6 marks)
benefitting the nervous system.

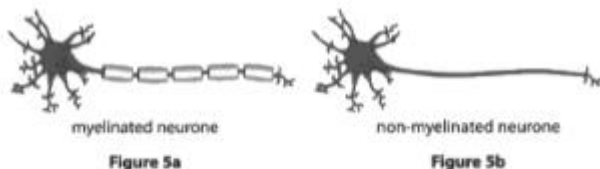
Lead Examiner comment: This response was awarded 0 marks.

Although the learner has attempted this question there is insufficient to be creditworthy.

The learner has the speeds the wrong way round and has described the myelinated neurone as obstructed, saying this would slow down the impulse. The statement that the non-myelinated neurone is "simpler" is too vague to be creditworthy, especially considering the rest of the learner's response. There is not adequate knowledge for Band 1.

5 Figure 5a shows the structure of a myelinated neurone.

Figure 5b shows the structure of a non-myelinated neurone.



Discuss the reasons for the differences in the speed of impulse conduction along myelinated and non-myelinated axons.

(6)

the non myelinated neurone structure is simpler meaning the speed of impulse conduction is quicker, the myelinated neurone has a structure in which impulses have more obstructions to go through slowing down the impulses.

Summary

Centres need to fully prepare learners for the exam by practicing exam technique, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question to reread their response in order to ensure that the question set has been addressed in the answer they have given, and that they have used appropriate scientific knowledge and vocabulary. Learners should have a clear understanding of the difference between “function” and “structure” and ensure they answer the question appropriately, for example if they are asked for the function of an organelle, they don’t just provide the name.

Based on their performance on this paper, learners should:

- Understand the demand of the command verbs, especially the difference between describe and explain
- Be familiar with and recognise the ultrastructure and function of organelles in the following cells: prokaryote cells (bacterial cells), eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific)
- Revise key definitions with Level 3 detail
- Show their working out for all calculations to ensure marks can be given for the stages within the calculation even if the final answer is incorrect
- Use the additional guidance to understand the depth and breadth of the specification

Individual Questions

Chemistry

Q1(a)(i)

It was based around the reactivity of a series of metals. In the first question learners were asked to give a use of silver that relies on its low reactivity. Learners performed well with the majority being able to give a correct use. The most common correct answer seen was jewellery or specified names of jewellery such as rings, earrings or necklaces, all of which gained credit.

Lead Examiner comment: *A less common answer seen was wiring, which was accepted, as in the case that gained the mark.*

(a) (i) Give **one** use of silver that relies on the low reactivity of the metal.

(1)

.....wires.....

Where learners lost marks, it was often where they were too vague with their answers.

Lead Examiner comment: *In this example the learner gained no mark as they describe a property rather than a use.*

(a) (i) Give **one** use of silver that relies on the low reactivity of the metal.

.....used as a conductor for electricity⁽¹⁾.....

Q1(a)(ii)

This question was also well answered with the majority of learners being able to name a metal that is more reactive than sodium.

Lead Examiner comment: *The most common answers were Francium and Rubidium.*

- (ii) Name **one other** metal, **not** shown in Figure 1, that is more reactive than sodium.

You may use the periodic table to help you answer the question.

(1)

~~Francium~~ Francium

Lead Examiner comment: *The most common incorrect answer seen that gained no credit was lithium.*

- (ii) Name **one other** metal, **not** shown in Figure 1, that is more reactive than sodium.

You may use the periodic table to help you answer the question.

(1)

Lithium

In some cases, although it was clearly stated in the question that it should be a metal that was not shown in Figure 1, learners have potassium as their answer. This did not gain credit. Learners should be taught to read the question stem carefully.

Q1(a)(iii)

Learners found the question more difficult. Many understood that calcium is in group 2 whereas sodium is in group 1, but they could not then relate that to the number of electrons in the outer shell of the atoms and therefore the ease of removing these electrons.

Lead Examiner comment: *This example scores both marks available.*

(iii) Explain, in terms of atomic structure, why calcium is less reactive than sodium.

(2)

Calcium is less reactive due to having to lose two outer shell electrons whereas sodium only needs to lose 1. This means more energy is required to remove the electrons from calcium, thus making it less reactive

Fewer learners took the alternative route of answering and stated that the calcium had more electrons, these often found it even harder to gain the second marking point and often scored just 1 mark out of the two available.

Q1(b)

It was generally well answered, many learners were able to state that copper is oxidised and silver is reduced to gain 2 marks, however often they found it harder to explain why.

Lead Examiner comment: Some learners gave a good explanation in terms of oxidation states as in this example which gained all 4 marks.

Explain which element is oxidised and which element is reduced in this reaction.

(4)

oxidised Copper in Cu has been ~~oxidised~~ or oxidation state of 0. In $\text{Cu}(\text{NO}_3)_2$, copper has been ~~oxidised~~ or oxidation state of +2. Therefore copper has been oxidised as there has been an increase in oxidation number. ~~It is by~~ 2

reduced Silver in AgNO_3 has an oxidation state of +1. In Ag, silver has an oxidation state of 0. Therefore silver has been reduced as there has been a ~~reduction~~ decrease in oxidation number by 1

Lead Examiner comment: *In some cases, learners got these the wrong way around, but with error carried forward could still gain marks for showing an understanding that oxidation is loss of electrons and reduction is gain of electrons. This example gained two marks.*

(b) When copper metal is placed in a solution of silver nitrate, a redox reaction takes place.

The equation for this reaction is:



Explain which element is oxidised and which element is reduced in this reaction.

(4)

oxidised silver has been oxidised
because it has lost electrons
due to the fact it's less reactive
than copper

reduced copper has been reduced
because it has gained electrons
due to the fact it is more
reactive than silver

Q2(a)(i)

Lead Examiner comment: *Learners started well with the majority being able to give a property of transition metals with the majority giving common properties of metals, such as good heat or electrical conductivity, high melting point or malleability as in this example that gained the mark.*

2 (a) Manganese is a transition metal.

(i) Give **one** property of transition metals.

(1)

They are good conductors of heat

Lead Examiner comment: *A common answer that did not score was where learners tried to explain a property in terms of incomplete d-subshells rather than giving a property, as in this example that did not score.*

2 (a) Manganese is a transition metal.

(i) Give **one** property of transition metals.

(1)

They dont have a complete 'd' subshell.

Q2(b)

This was the first calculation on the paper. Learners were asked to calculate the concentration of solution of potassium permanganate.

Lead Examiner comment: *Many answered this well, by setting out their answer in a logical way, dividing the mass by the volume and scoring the full 2 marks.*

(b) Potassium permanganate, KMnO_4 , is a compound of manganese.

0.050 dm³ of a solution of potassium permanganate contains 1.975 g of potassium permanganate.

Calculate the concentration of this solution in g dm⁻³.

(2)

$$\text{concentration} = \frac{\text{amount}}{\text{volume}}$$

$$\frac{0.050}{1.975} = 0.025$$

$$\text{concentration} = 0.025 \text{ g dm}^{-3}$$

Q2(c)(i) and Q02(c)(ii)

Learners were asked to calculate the number of mols of potassium ions in a solution and from that the number of ions by multiplying by Avogadro's constant using the formula provided. The majority of learners were able to calculate the number of moles to gain the mark.

An alternative number of moles was given in the question for those that could not answer Q2(c)(i) or who were unsure of their answer. It was pleasing to see that many learners unitised this and were still able to gain marks in part Q2(c)(ii) when they missed Q2(c)(i) or were unsure of their answer.

Lead examiner comment: *In this example the learner correctly calculated the number of moles and rounded it correctly which gained the mark in Q2(c)(i). They then used the given value to calculate the number of ions in the sample to gain the full 2 marks in Q2(c)(ii).*

(c) (i) The solution contains 0.489g of potassium ions.
Calculate the number of moles of potassium ions in the solution. (1)

relative atomic mass of potassium = 39.1

$$\frac{0.489}{39.1} = 0.0125$$

number of moles = 0.013

(ii) Calculate, using your answer to (c) (i), the number of potassium ions in the solution.
If you did not get an answer for (c) (i), use the value 0.016 for the number of moles. (2)

Use the equation:
number of ions = number of moles $\times$ Avogadro constant
Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$
Show your working.

$$0.016 \times 6.02 \times 10^{23} = 9.632 \times 10^{21}$$

number of potassium ions = 9.632×10^{21}

(Total for Question 2 = 7 marks)

Lead Examiner comment: *In this example, the learner multiplied the mass by the relative atomic mass and so lost the mark in Q2(c)(i), they took this answer into their Q2(c)(ii) and with error carried forward gained both marks for multiplying their number of moles by Avogadro's constant.*

(c) (i) The solution contains 0.489g of potassium ions.

Calculate the number of moles of potassium ions in the solution.

(1)

relative atomic mass of potassium = 39.1

$$0.489 \times 39.1$$

$$= 19.1199$$

number of moles = 19.1199 g/m

(ii) Calculate, using your answer to (c) (i), the number of potassium ions in the solution.

If you did not get an answer for (c) (i), use the value 0.016 for the number of moles.

(2)

Use the equation:

number of ions = number of moles $\times$ Avogadro constant

Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$

Show your working.

$$19.1199 \times 6.02 \times 10^{23} \text{ mol}^{-1}$$

$$= 1.1509638 \times 10^{25}$$

Lead Examiner comment: *In the following example, the learner has correctly calculated the number of moles of potassium ions in Q2(c)(i), it can be seen that this was taken forward and used correctly in Q2(c)(ii) but unfortunately when taking the answer on to the answer line the learner has omitted the $\times 10^{23}$ and so loses the mark. This exemplifies the importance of showing working. If the learner had not shown their working and just written 7.53 on the answer line, the first marking point could not have been awarded.*

(c) (i) The solution contains 0.489g of potassium ions.
Calculate the number of moles of potassium ions in the solution. (1)

relative atomic mass of potassium = 39.1

$$\frac{M}{m/n} = 0.01250638$$

number of moles = 0.0125g

(ii) Calculate, using your answer to (c) (i), the number of potassium ions in the solution.
If you did not get an answer for (c) (i), use the value 0.016 for the number of moles. (2)

Use the equation:
number of ions = number of moles $\times$ Avogadro constant
Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$
Show your working.

$$0.0125 \times 6.02 \times 10^{23} = 7.5288$$

number of potassium ions = 7.53

(Total for Question 2 = 7 marks)

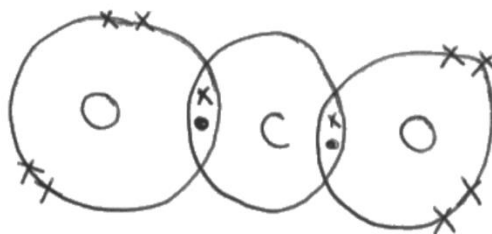
Q3(b)

In general, learners found drawing the dot and cross diagram in Q3(b) quite difficult with many losing marks.

Lead Examiner comment: *The most common error was to draw just one shared pair between the carbon and oxygen atoms. This gained no credit.*

(b) Draw the dot and cross diagram for a molecule of carbon dioxide.

(2)



Other errors included trying to draw ionic bonds or when correctly drawing the double bonds then adding further electrons to the carbon, leaving it with 10 electrons or drawing carbon monoxide rather than carbon dioxide.

Learners often drew the carbon dioxide without 'shells' - this is fine and gained the marks.

Lead Examiner comment: *This example gained 1 mark for the two shared pairs on electrons between the carbon and each oxygen. Unfortunately, they missed the extra electrons on each oxygen and did not gain the second mark for the rest of the molecule correct.*

(b) Draw the dot and cross diagram for a molecule of carbon dioxide.

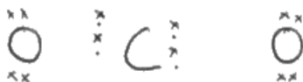
(2)



Lead Examiner comment: *This last example gained 2 marks.*

(b) Draw the dot and cross diagram for a molecule of carbon dioxide.

(2)



Q3(c)

This was very poorly answered across all papers with few scoring full marks. In some cases, learners gained 1 mark for showing an understanding that there were 4 hydrogens around the central carbon atom or that the methane is tetrahedral. Very few referred to there being 4 electron pairs or bonds around the carbon.

Lead Examiner comment: *This example scored 1 mark for stating that the carbon is bonded to 4 hydrogens. The learner states that the hydrogen are equally spaced out but not say why this is so and so does not gain the second mark.*

Explain why all the bond angles in methane are 109.5° .

(2)

because one carbon is bonded to 4
hydrogens and they are all equally
spaced out in all direction

Lead Examiner comment: *Only the strongest learners referred to the electron pairs, bonds or hydrogen atoms repelling to gain the second marking point. The example scored 2 marks.*

Explain why all the bond angles in methane are 109.5° .

(2)

Tetrahedral, all bonds are 109.5 because
 there are no lone pairs and 4 bond pairs,
 which all repel equally, since bond pair want to
 repel each other

Q3(c)

Learners also found this question difficult. Learners that did well often showed working around their answers.

Lead Examiner comment: *In this example, the learner knew that hydrogen bonding is only found in the ammonia, whereas permanent dipole-dipole forces are found in ammonia, hydrogen chloride and methane and temporary induced dipole forces are found in all of the molecules.*

(d) Simple molecular compounds have weak intermolecular forces between the molecules.

Complete Table 1 with ticks and crosses to show which intermolecular forces exist between the molecules.

Carbon dioxide has been completed for you.

(3)

molecule	structure	intermolecular force		
		hydrogen bonding	permanent dipole - dipole forces	temporary - induced dipole forces
carbon dioxide	$\text{O}=\text{C}=\text{O}$	x	x	✓
ammonia	$\begin{array}{c} \text{H}-\text{N}-\text{H} \\ \\ \text{H} \end{array}$	✓	✓	✓
hydrogen chloride	$\text{H}-\text{Cl}$	x	✓	✓
methane	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$	x	x	✓

Table 1

Q4

The last question on the paper was the six-mark extended open responses with a levels-based mark scheme. This question differentiated well with less able learners making simple or generic statements with few linkages and application of knowledge and stronger learners demonstrating a more comprehensive knowledge and understanding and applying the relevant knowledge to the question.

Lead Examiner comment: *In this example, the learner scored 2 marks in Level 1. The learner gives some basic ideas and shows an understanding that the ionic bonding is between positive and negative ions and that there are strong attractions between these ions. They have the trend correctly understanding that the lower the melting point the bonds are easier to break, however there is no explanation in terms of charges on ions or ionic radii of the ions involved.*

- 4 Sodium fluoride, magnesium fluoride and magnesium chloride are all ionic metal halides.

Table 2 shows their formulae and melting points.

metal halide	formula	melting point (°C)
sodium fluoride	NaF	993
magnesium fluoride	MgF ₂	1261
magnesium chloride	MgCl ₂	714

Table 2

Explain the melting points of the **three** metal halides.

(6)

and ionic bonding is attraction between negative ions + positive ions.

The sodium fluoride the melting point increases ~~but~~ because it has a strong attraction between the positive ions and the negative ions and you can't break the bond. Whereas magnesium fluoride has an even higher melting point because the attraction between the two must be higher making it harder for the bonds to break.

The magnesium chloride has a less higher melting point because the attraction between the positive ions + the negative ions must be a little bit weaker making the bonds easier to break.

Lead Examiner comment: In this example, the learner scored 4 marks in Level 2. They understand that magnesium chloride has the weakest electrostatic attraction and links this to the lowest melting point. The link to weakest electrostatic attraction and lowest melting point is good, but there are issues in the use of scientific terms with reference to biggest atoms rather than ions and lack of clarity as to which ion is the biggest. This answer is moving towards a Level 3 answer, but lack of clarity regarding the size of ionic radii or the charges on the ions gained a mark of 4 in Level 2.

- 4 Sodium fluoride, magnesium fluoride and magnesium chloride are all ionic metal halides.

Table 2 shows their formulae and melting points.

metal halide	formula	melting point (°C)
sodium fluoride	NaF	993
magnesium fluoride	MgF ₂	1261
magnesium chloride	MgCl ₂	714

Explain the melting points of the three metal halides.

Sodium fluoride, magnesium fluoride and magnesium chloride are all ionic metal halides. Magnesium chloride contains the largest atoms of the three and therefore has the weakest electrostatic forces of attraction due to shielding, resulting in a lower melting point. Magnesium fluoride may have larger atoms than sodium fluoride but it has more bonds (2F for every 1 Mg) resulting in the bonds being harder to break so a higher melting point. Sodium fluoride has the slightly smaller atoms than magnesium fluoride but less bonds giving it a lower melting point. It is smaller than magnesium chloride and so has a higher melting point.

Summary

Based on their performance on this paper:

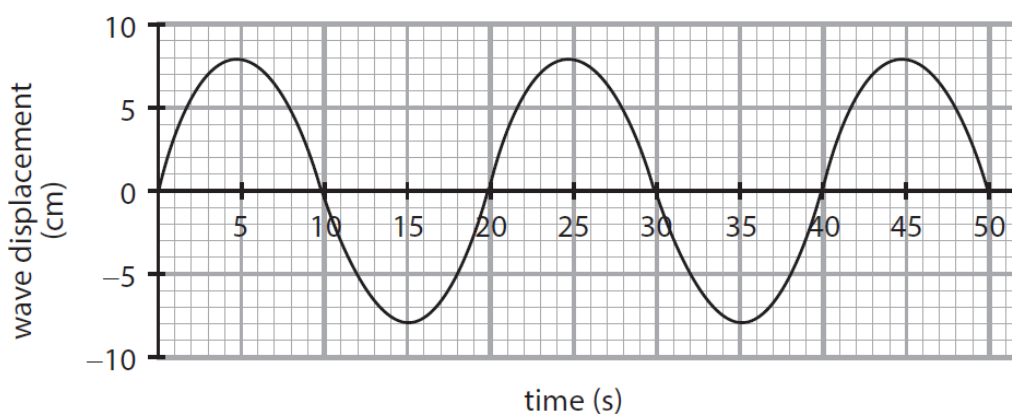
- Learners should practice carrying out calculations from the unit, setting out and showing their workings in a logical manner so that they can arrive at a given answer in a concise way
- Learners should continue to practice exam technique, ensuring that they know what is required by specific command words and know the difference in what is required from commands such as describe and explain and calculate and state
- Learners should take care that they use key subject carefully and in the correct way for example the use of ions that atoms
- Centres should ensure that learners understand that they should not give lists with more than one answer for a question unless specifically asked to do so

Individual Questions

Physics

Q1(a)

All the information for the answers to Q1(a)(i), Q1(a)(ii) and Q1(a)(iii) were obtained from the diagram.



More than half of learners were able to give correct answers to all parts of this question. Learners needed to note the labels on the axes and the scales.

- For Q1(a)(i) the graph shows $2\frac{1}{2}$ oscillations
- The amplitude Q1(a)(ii) is the maximum displacement from the line of zero displacement. The vertical scale which is given in centimetre (cm) shows this to be 8 cm
- The period for Q1(a)(iii) is the time for one complete oscillation which is read from the x axis and is 20 sec

Q1(b)

This question was more demanding as the motion of particles in a transverse wave had to be recalled and then applied to water molecules.

Lead Examiner comment: *Both answers below gained 2 marks. The first mark was for giving that the water molecules oscillate, vibrate or move up and down and the second mark for knowing that this the motion of the particles is perpendicular to the direction of travel (of the wave) or the idea that the water molecules do not move in the direction of the wave.*

(b) The surface water wave is a transverse wave.

Describe how the water molecules move in the wave.

(2)

water molecules move in oscillations perpendicular to the direction of travel. This is due to the wave being transverse

(b) The surface water wave is a transverse wave.

Describe how the water molecules move in the wave.

(2)

The water molecules vibrate on the spot transferring energy

Q1(c)

The majority of learners were able to gain 2 marks for this type of calculation. The equation is given as is the meaning of the symbols used. Learners should start by substituting in the equation given, if the rearrangement is then incorrect a mark can still be awarded for the evaluation.

Lead Examiner comment: *The response shown below shows the correct substitution, rearrangement and evaluation.*

(c) The speed (v) of a different surface water wave is 1.5 m s^{-1} .

The frequency (f) of the wave is 2.5 Hz .

Calculate the wavelength (λ) of the wave.

(3)

Use the equation: $v = f \lambda$

Show your working.

$$\begin{aligned} v &= f \lambda & \lambda &= \frac{v}{f} & \frac{1.5}{2.5} \\ 1.5 &= 2.5 \lambda & & & \\ & & & & = 0.6 \end{aligned}$$

wavelength = 0.6 m

A common error which meant marks were lost was in not rounding the final answer correctly. With incorrect rearrangement some learners arrived at $2.5/1.5$ this gives the answer 1.666 recurring or can be rounded to 1.67. Leaving the answer as 1.6 in the answer line did not get the evaluation mark as it was incorrectly rounded.

Q2(a)

Although learners should be familiar with diffraction gratings very few could describe that it was the 'spreading out' of waves as the waves pass through small gaps in a diffraction grating that allows emission spectra to be produced.

Q2(b)

The majority of learners were able to identify the gases that were present in the atmosphere of the star and many were able to give an accurate reason for their choice by saying that the spectral lines of gases P and S matched the lines shown in the atmosphere of the star

Lead Examiner comment: *The response shown below gained 3 marks.*

(b) The atmosphere of a star contains two gases.

Figure 2 shows the emission spectra of four gases, P, Q, R and S, and the emission spectrum of the light from the atmosphere of the star.

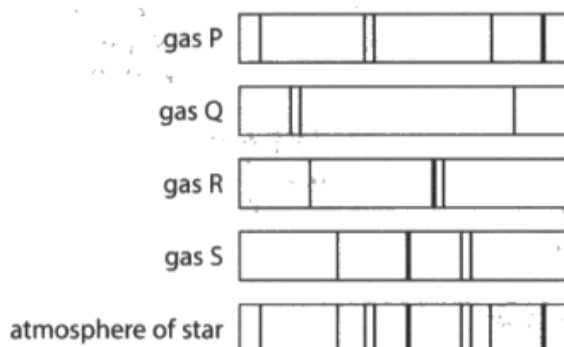


Figure 2

(i) Identify the two gases in the atmosphere of the star, using information from Figure 2.

(1)

1 gas P

2 gas S

(ii) Give two reasons for your choices in (b)(i).

(2)

1 gas P shows in the atmosphere of the star as the lines match to the results as well as the lines in gas S.

2 gas Q and gas R do not appear in the findings of the atmosphere of star, as their emission spectra does not match the emission spectrum of the star.

Q3(a)

Only about half of learners could select B as the correct answer to this multiple-choice question. Learners should be familiar with the standard way of drawing a 'normal' at 90° to an interface as all angles in optics are measured from the normal.

3 Figure 3 shows refraction at an air-glass boundary.

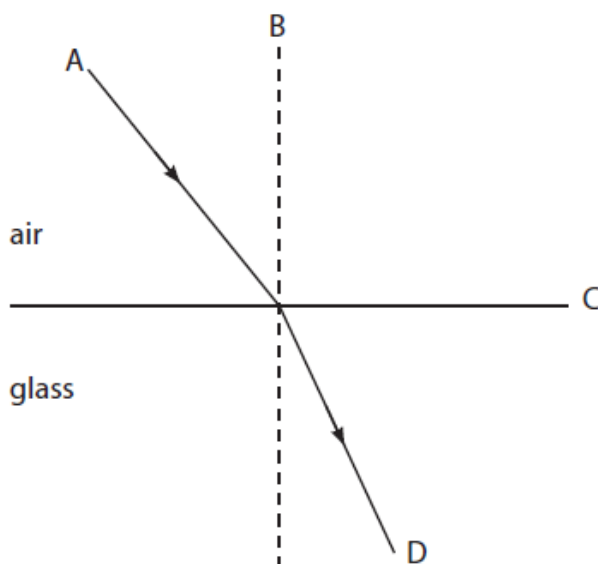


Figure 3

(a) Which line in Figure 3 is the normal?

Q3(b)

This question led to the full range of marks being awarded. A significant number of learners were able to deal with 'sin' and ' $\sin^{-1}$ ' to get the correct answer of (29.7) 30° as shown in the response below.

However, for a few learners more practice is needed with the use of this equation as substitution could not be completed, the 'sin' in the equation was ignored and the compensatory mark for giving $c/v = 1.5$ was all that could be awarded.

The mark quite frequently awarded was 3 where the learner obtained the value for $\sin r$ but could then not use their calculator to find the angle r

Lead Examiner comment: *This response gains 4 marks, angle r determined.*

(b) A different ray of light has an angle of incidence (i) of 48° .

The speed of light in air (c) is $3.0 \times 10^8 \text{ m s}^{-1}$.

The speed of light in glass (v) is $2.0 \times 10^8 \text{ m s}^{-1}$.

Calculate the angle of refraction (r) for this ray of light.

(4)

Use the equation:

$$\frac{c}{v} = \frac{\sin i}{\sin r}$$

Show your working.

$$\frac{3.0 \times 10^8}{2.0 \times 10^8} = 1.5$$

$$1.5 = \frac{\sin(48)}{\sin r}$$

$$\sin r = \frac{\sin(48)}{1.5} = 0.495$$

$$\sin^{-1}(0.495) = 29.66$$

$$\text{angle of refraction } (r) = 29.7^\circ$$

Lead Examiner comment: *This response gains 3 marks, sin r determined not converted to an angle.*

The speed of light in air (c) is $3.0 \times 10^8 \text{ m s}^{-1}$.

The speed of light in glass (v) is $2.0 \times 10^8 \text{ m s}^{-1}$.

Calculate the angle of refraction (r) for this ray of light.

(4)

Use the equation:

$$\frac{c}{v} = \frac{\sin i}{\sin r}$$

Show your working.

$$\frac{3.0 \times 10^8}{2.0 \times 10^8} = \frac{\sin 48}{\sin r}$$

$$1.5 = \frac{\sin 48}{\sin r}$$

$$\sin r = \frac{\sin 48}{1.5}$$

$$= 0.5$$

angle of refraction (r) = 0.5 °

Q4(a)(i)

This question was the most demanding as many learners confuse the analogue and digital signals with their graphical representation. The labels on the axes of the sketch graph must be read carefully as it shows that the 'signal voltage' (y axis) varies with 'time' (x axis).

Lead Examiner comment: *The example below gained both marks.*

Figure 4a shows an analogue signal.

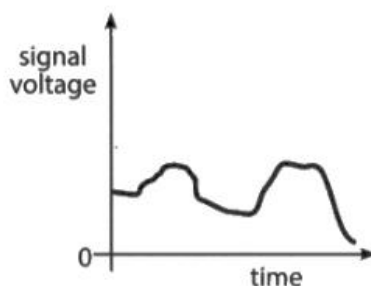


Figure 4a

(a) (i) Describe the analogue signal shown in Figure 4a.

(2)

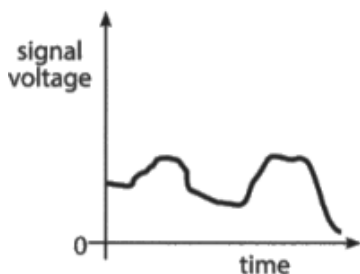
as we can see on the figure the
signal voltage varies in voltage, also as
time goes on at the end ^{the signal} it decreases
massively.

Q4(a)(ii)

Many learners did not use the information in the three graphs and relate these as an example of analogue-digital conversion with differing sampling rates.

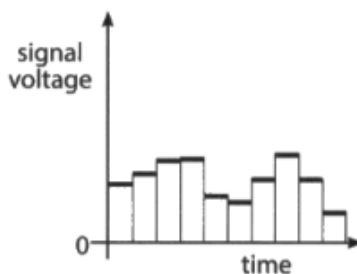
- (ii) An analogue-digital converter is used to convert the analogue signal to a digital signal.

The first stage in the conversion process is to sample the signal.



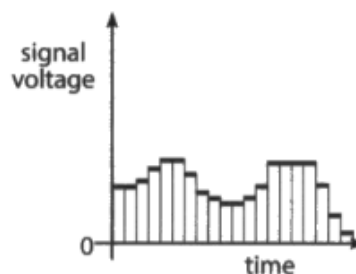
analogue signal

Figure 4a



digital signal
low sampling rate

Figure 4b



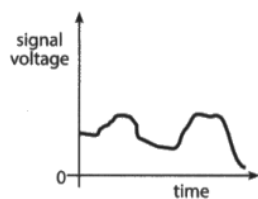
digital signal
higher sampling rate

Figure 4c

Lead Examiner comment: *This response gained two marks by referring to the lower sampling rate producing a less accurate version of the original analogue signal.*

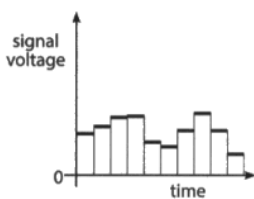
- (ii) An analogue-digital converter is used to convert the analogue signal to a digital signal.

The first stage in the conversion process is to sample the signal.



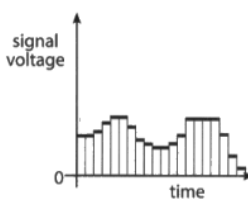
analogue signal

Figure 4a



digital signal
low sampling rate

Figure 4b



digital signal
higher sampling rate

Figure 4c

Figure 4b shows the digital signal produced when using a low sampling rate.

Figure 4c shows the digital signal produced when using a higher sampling rate.

Explain the importance of sampling rate in the conversion of analogue signals to digital signals.

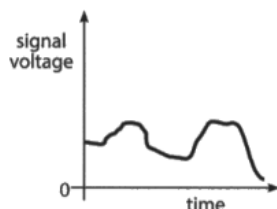
(2)

Sampling rate is important because a lower digital signal made with a low sampling rate will have be of lower quality as the are less accurate to the origal analogue signal

Lead Examiner comment: *This response gains both marks by referring to the higher sampling rate producing a digital signal which is closer to the original analogue signal.*

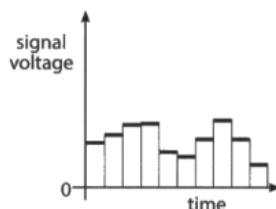
(ii) An analogue-digital converter is used to convert the analogue signal to a digital signal.

The first stage in the conversion process is to sample the signal.



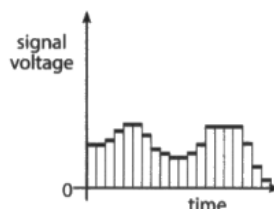
analogue signal

Figure 4a



digital signal
low sampling rate

Figure 4b



digital signal
higher sampling rate

Figure 4c

Figure 4b shows the digital signal produced when using a low sampling rate.

Figure 4c shows the digital signal produced when using a higher sampling rate.

Explain the importance of sampling rate in the conversion of analogue signals to digital signals.

(2)

A higher sampling rate allows for a better a higher number of discrete values to assign each point on the analogue signal, allowing it to be closer to the original analogue signal.

Q4(b)

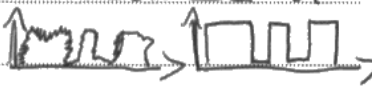
Learners were asked to explain one advantage of using digital signals rather than analogue. The most common correct answer was that there was less interference with digital signals, and this got the first mark. However only a few learners could get the second mark and explain that it was due to digital using binary code.

The most common misconception was that digital signals travel faster, digital and analogue travelling through the same medium travel at the same speed. However, digital signals can travel further because they can be regenerated. Similarly digital signals are more secure because they can be encrypted.

Lead Examiner comment: *To gain both marks there must be an identification point and an expansion as below.*

(b) Explain **one** advantage of using digital signals rather than analogue signals.

(2)

digital signals can pick up interference
but that can be cleared up because it
can be only one's and zeros 

Q5(a)

The majority of learners knew that Wi-Fi can be used to connect multiple devices to the internet.

Q5(b)

The extended open response was to compare signals transmitted to a home using optical fibres or satellites.

The majority of responses concentrated on the differences between optical fibres and satellite transmission with very few references to similarities. Learners need to present their answers so that they are coherent and logical. It is often difficult to find comparisons when they are scattered through two or more pages. Learners also need to focus on what the question asks about. Medical uses of optical fibres was not relevant and not credited. The answers are marked using the indicative content as a guide to the content that might be expected and the mark descriptors which consider not only content but lines of argument and the structure and coherence of the response.

Lead Examiner comment: *Level 1, 2 marks.*

The example below has a correct comparison. Optical fibres use a 'higher wave frequency' which is sufficient to achieve Level 1. The remaining incorrect information about optical fibres being in the Earth's atmosphere is ignored. There is also a correct fact that 'satellites are based outside the Earth's atmosphere'.

(b) Signals can be transmitted to a home using optical fibres or using satellites.

Compare the transmission of signals using optical fibres with the transmission of signals using satellites.

Your answer should include similarities and differences of the:

- method of transmission
- wave frequency.

(6)

Optical Fibers is in the earth's
atmosphere meaning the wave
frequency is higher
satellites are based out of the
earth's atmosphere meaning the
wave frequency is lower.

Lead Examiner comment: *Level 2, 4 marks. This example shows some good knowledge, relevant information and has a clear logical structure.*

(b) Signals can be transmitted to a home using optical fibres or using satellites.

Compare the transmission of signals using optical fibres with the transmission of signals using satellites.

Your answer should include similarities and differences of the:

- method of transmission
- wave frequency.

(6)

satellites ^{use} ~~use~~ radio waves to transmit signals because of their ~~wavelength~~ long wavelength and ^{low} ~~short~~ frequency while optical fibres use visible light which have a ~~short~~ wavelength but higher frequency.

Lead Examiner comment: *Level 3, 6 marks. This response shows a logical, coherent series of comparison, in a logical chain of reasoning. It is a sustained application of relevant evidence to support the comparison.*

(b) Signals can be transmitted to a home using optical fibres or using satellites.

Compare the transmission of signals using optical fibres with the transmission of signals using satellites.

Your answer should include similarities and differences of the:

- method of transmission
- wave frequency.

(6)

Fibre optic cables use small ~~thin~~ fibres of glass and ~~placed~~ ^{are placed} under water with light passing through to send signals. This contrasts with using satellites because satellites use radiowaves to send signals and microwaves to send signals to outer space ~~it's~~ ⁱⁿ orbiting satellites. Albeit the frequency of light is much higher than radio and microwaves, it doesn't go in a straight line but bounces off of the walls of the fibre optic cable.

Summary

Points which should be noted to help learners to improve:

- Use the information given on graphs, take note of the units and axes labels
- Always show substitution into an equation then carry out the rearrangement
- Learn to describe how particles behave in transverse and longitudinal waves
- Practice using calculator to convert an angle to the sin of the angle
- Practice using $\sin^{-1}$ to find an angle
- Practice drawing and labelling ray diagrams for refraction
- Learn how sampling rate affects analogue to digital conversion



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